

HireStream — Infrastructure Provisioning Request

Hardware · Software · Network · Access — Staging & Production · HPSEDC

This lists the **HireStream-specific** hardware, software, network and access requirements for deployment on government infrastructure (**Staging + Production**). It assumes the IT team's standard policies — OS hardening, NTP, logging, base monitoring, firewall hygiene — are already in place; only items specific to this application are stated.

Air-gapped. The VMs are isolated from the internet, accessed via SSH through a Windows jump host. So the software in §2 must be **pre-installed (or available on an internal package mirror)**; HTIS delivers the application and its dependencies as a **pre-built artifact via the jump host** — no online build runs on the server.

Assumptions — please confirm: HP State Data Centre hosting; PROD sized to the FRS target of 5,000 concurrent users; one VM per environment (application + PostgreSQL co-located); 99.9% uptime via a single VM + standard backups (a standby DB replica can be added if a penalty-backed SLA applies).

1. Hardware

Resource	Staging (STG)	Production (PROD)
vCPU	4	8
RAM	8 GB	16 GB
Disk (SSD)	100 GB	200 GB (expandable)
Apps hosted	HireStream + Verify + DB	HireStream + DB

Separate VMs. Candidate document uploads grow disk over time — please allow online disk expansion, or provide object storage.

2. Software to pre-install / stage on disk (air-gapped — exhaustive)

As the server has **no internet access**, every item below must be **pre-installed, or staged on local disk / an internal apt mirror**, before handover. Versions match our dev & staging environment.

Category	Software	Version	Purpose
Operating system	Ubuntu Server LTS	24.04 LTS (Noble)	Base OS — same as our other deployments
Base & build	build-essential (gcc, g++, make), python3	distro	Native module builds (e.g. bcrypt)
Base & build	git, curl, wget, unzip, tar, rsync	distro	Deployment / artifact transfer
Base & build	ca-certificates, openssl	distro	TLS / crypto
Runtime	Node.js	20.x LTS (20.20+)	Application runtime
Runtime	PM2 (global npm)	6.x	Process manager — clustering + auto-restart
Database	PostgreSQL server + client + contrib	16.x (16.14+)	Primary database

Category	Software	Version	Purpose
Web / proxy	Nginx	1.24+	TLS reverse proxy, static serving, rate limiting
Optional	Redis	7.x	Session store / cache

PM2 is a global npm package — it (and the build tools) must also be available **offline**. Our application's own npm dependencies are bundled into the delivered artifact, so those are **not** needed from IT.

3. Network & integration egress

Ports: 443 (HTTPS in) · 80 → 443 redirect · 22 (SSH, **from jump host only**) · Node ↔ PostgreSQL on localhost.

Internet egress is blocked, but the portal needs **outbound reachability to these government services** (via the internal network or a firewall exception) — otherwise the integrations cannot function:

- HIM Parivar / HIM Access SSO · Aadhaar / UIDAI · DigiLocker · Government Email (SMTP) · Government SMS gateway

Please advise which are reachable on the internal government network vs require firewall exceptions.

4. TLS certificates

Air-gapped, so Let's Encrypt / ACME is not possible. Please provide CA-issued certificates (full chain + private key) for:

- **PROD** domain: · **STG** domain:

5. Access required

#	Access	Detail
1	SSH via jump host	Named service account with sudo (app, PM2, Nginx)
2	File transfer via jump host	SCP/SFTP path for the app artifact + updates
3	PostgreSQL	Database + role able to create schema / extensions
4	Backup target	Location for DB dumps + VM snapshots

6. Deployment & operations

HTIS builds the release on a matching platform → transfers it via the jump host → runs database migrations → restarts under PM2 (rollback = previous artifact + DB snapshot). Daily PostgreSQL dump + WAL archiving to the backup target supports the 99.9% target.

Checklist for the IT team

- ☐ STG + PROD VMs provisioned (4 / 8 / 100 · 8 / 16 / 200)
- ☐ Ubuntu 24.04 LTS + Node 20 + PM2 + PostgreSQL 16 + Nginx (pre-installed or staged offline)
- ☐ TLS certificates provided for both domains
- ☐ Firewall: 443 / 80 / 22 + integration egress
- ☐ SSH service account + SFTP path + PostgreSQL database & role
- ☐ Backup target allocated